

CatGen

Comparing DCGAN, AAE, and VQ-VAE on cat image generation

Adam Kaniasty · Deep Learning Project III · June 2026

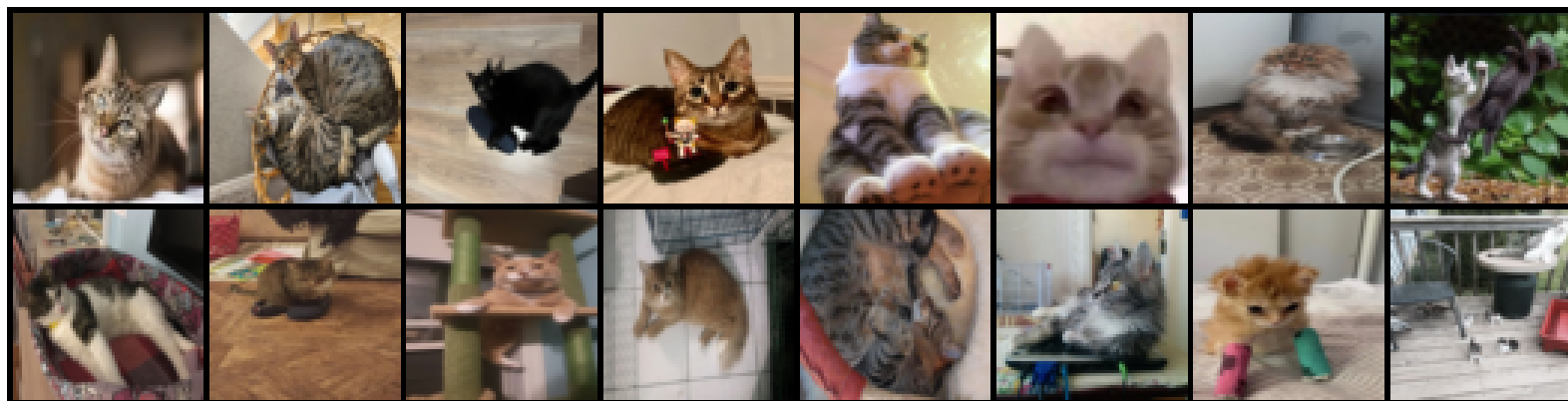
Goal

Question: Which lightweight generative model produces the best cat images under a fixed GPU budget?

- Three families: **DCGAN**, **Adversarial Autoencoder (AAE)**, **VQ-VAE**
- Metrics: **FID** + sample grids + latent interpolation (DCGAN/AAE)
- Extension: **cats + dogs** training for the **best** model only (128×128 phase)
- Scope: custom CNN-scale models, not StyleGAN or diffusion

Dataset

- **Main task:** generate cat images from the Kaggle Cat Dataset (~10k available images)
- **Part A:** 1500 train cats, 500 held-out reference cats for FID
- **Part B/C:** 3000 train cats, 1000 held-out reference cats for FID
- **Extension:** Dogs vs Cats data used only for the mixed cats+dogs DCGAN run
- **Preprocessing:** resize/crop → RGB → normalize to $[-1, 1]$; horizontal flip during training



Training cats after preprocessing

Models

Model	Generation path	Latent	Params
DCGAN	random $z \rightarrow$ generator $\rightarrow$ image; discriminator judges real/fake	Continuous z	~9.6M
AAE	image $\rightarrow$ encoder $\rightarrow z \rightarrow$ decoder; latent discriminator regularizes z	Continuous z	~7.7M
VQ-VAE	image $\rightarrow$ encoder $\rightarrow$ codebook $\rightarrow$ decoder; generation samples code indices	Discrete codes	~0.5M

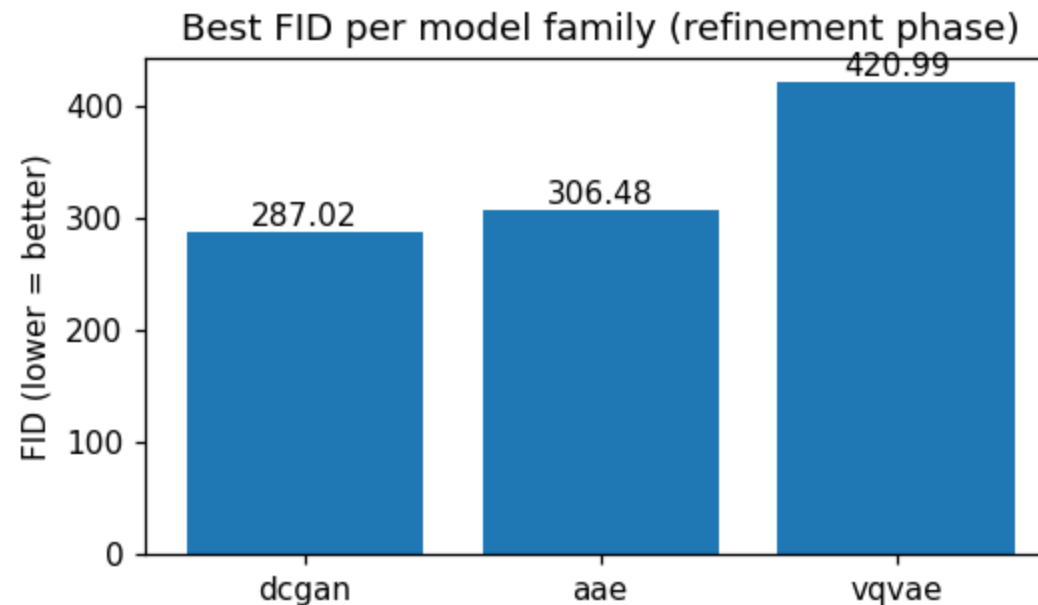
Training objectives

Model	Optimized losses	Purpose
DCGAN	discriminator BCE + generator non-saturating BCE	generate images that look real to D
AAE	reconstruction MSE + latent discriminator BCE + encoder adversarial BCE	reconstruct images and keep z sampleable
VQ-VAE	reconstruction MSE + vector-quantization commitment loss	reconstruct through a stable discrete codebook

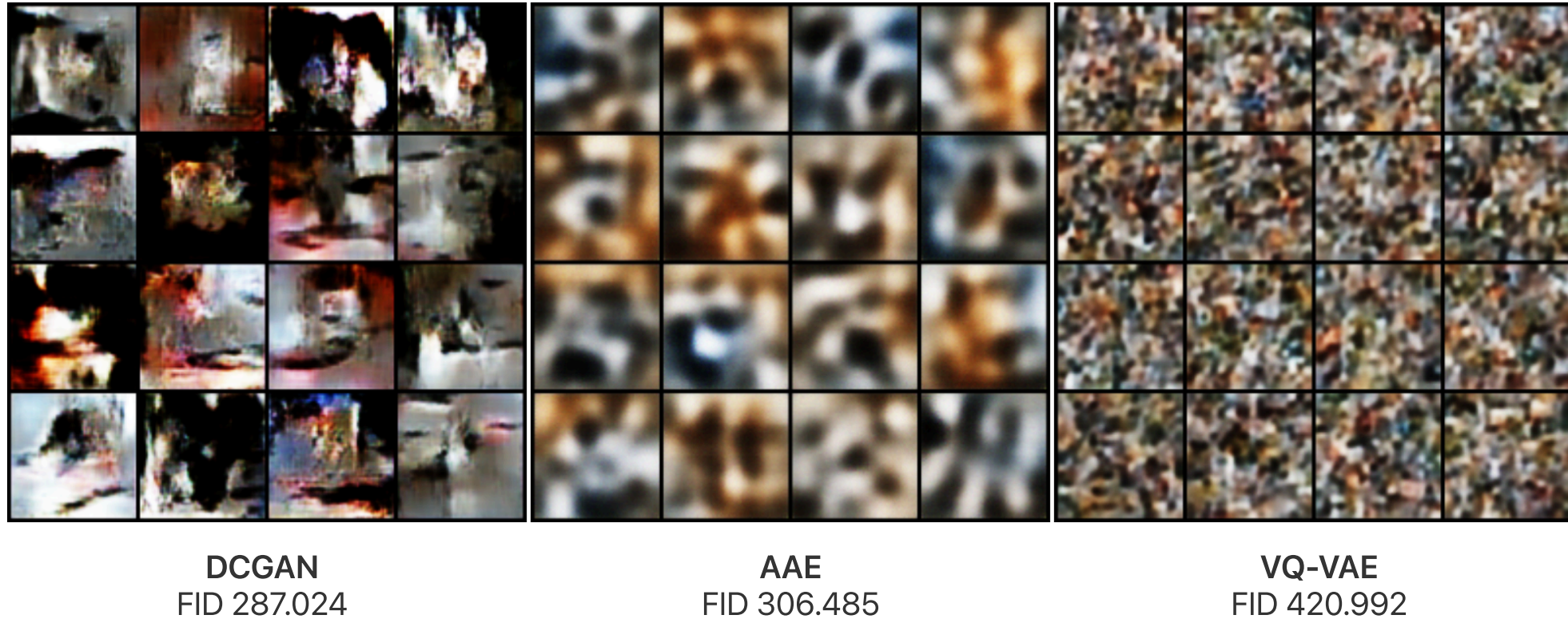
- DCGAN uses label smoothing on real discriminator targets in selected runs
- AAE regularizes the encoded latent distribution toward a Gaussian prior
- VQ-VAE generation samples discrete codes from empirical training-code frequencies

Part A — What we tried (64×64)

- **Sweep:** ~10 configs per family (z, lr, codebook K) on **1500** train cats
- Varied: latent size, learning rate, codebook size, generator upsampling, label smoothing
- **Refine:** 80 epochs, best configs per family; DCGAN used upsample+conv G and separate G/D LR
- Takeaway: broad sweep identified DCGAN as best FID model, but visual quality stayed weak



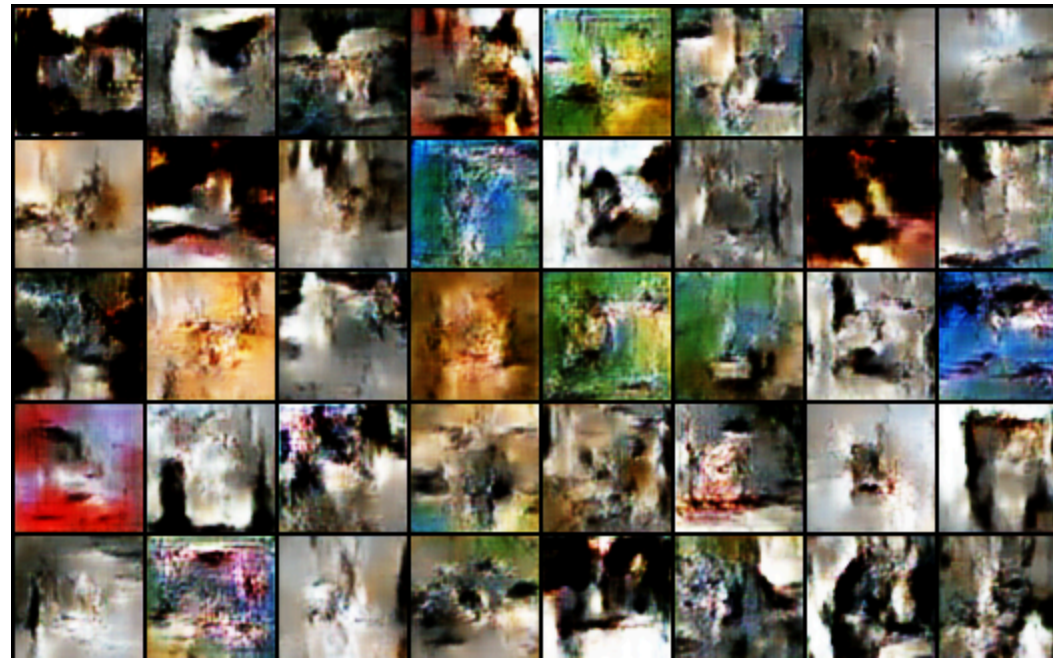
Part A — Sample grids (64×64, eval)



Part A — Outcome

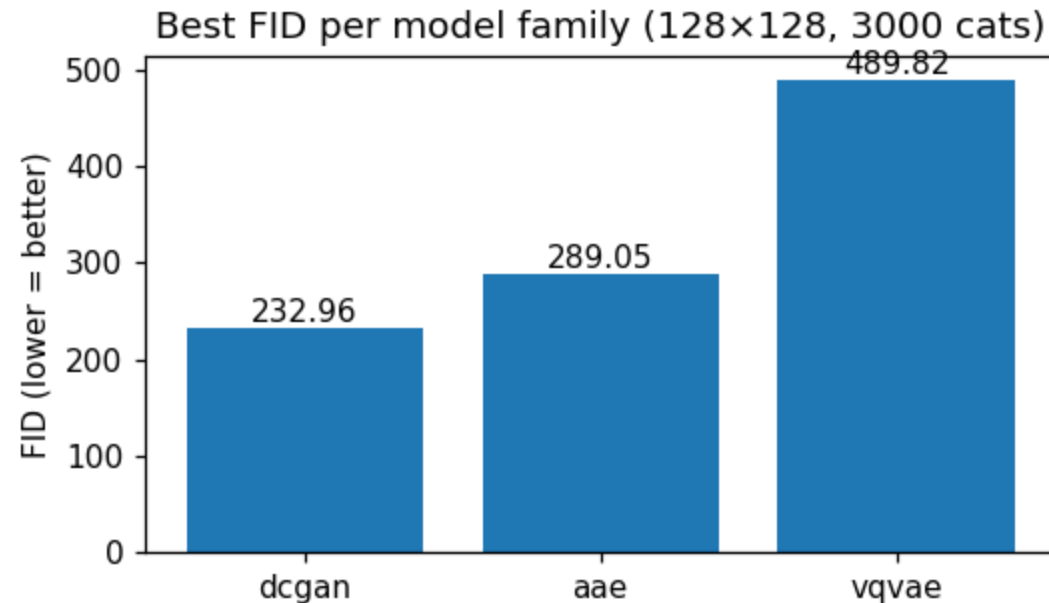
Model	Best refined FID ↓	Visual outcome
DCGAN	287.024	most structured, still noisy
AAE	306.485	smoother, less realistic
VQ-VAE	420.992	weakest distribution match

- **Conclusion:** 64×64 was useful for model selection, but not enough for convincing cat images

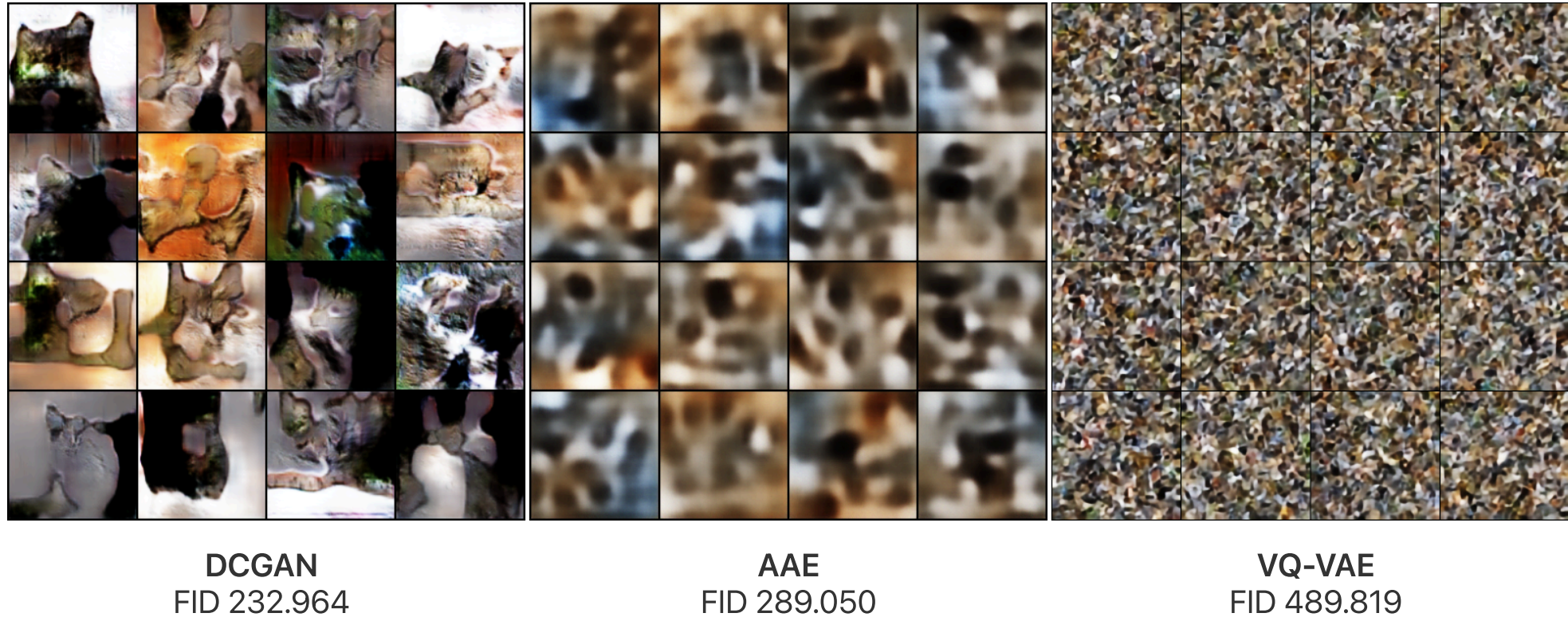


Part B — 128×128, 3000 cats

- Models retrained at 128×128: DCGAN, AAE, VQ-VAE
- Training split: 3000 cat images
- Training augmentation: RandomResizedCrop + horizontal flip
- Evaluation: FID computed against 1000 held-out reference cats
- Takeaway: increasing resolution and data improved DCGAN most clearly



Part B — Sample grids (128×128, eval)



Part B — DCGAN @ 128×128

Best 128×128 result: DCGAN, FID 232.964

Samples become more cat-like than at 64×64, but faces and textures remain distorted.

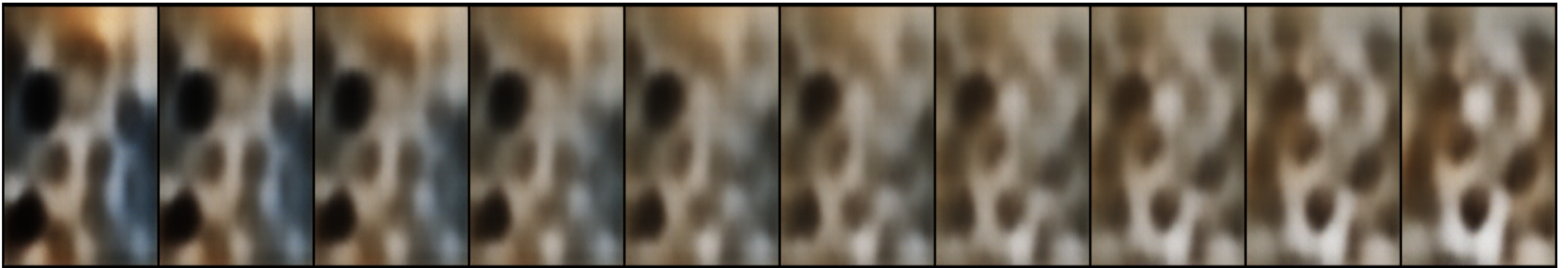


Part B — Latent interpolation

Interpolation checks whether the latent space changes smoothly; it is not a substitute for FID or sample quality.



DCGAN @ 128×128



AAE @ 128×128

Part B — Cats vs dogs (best model)

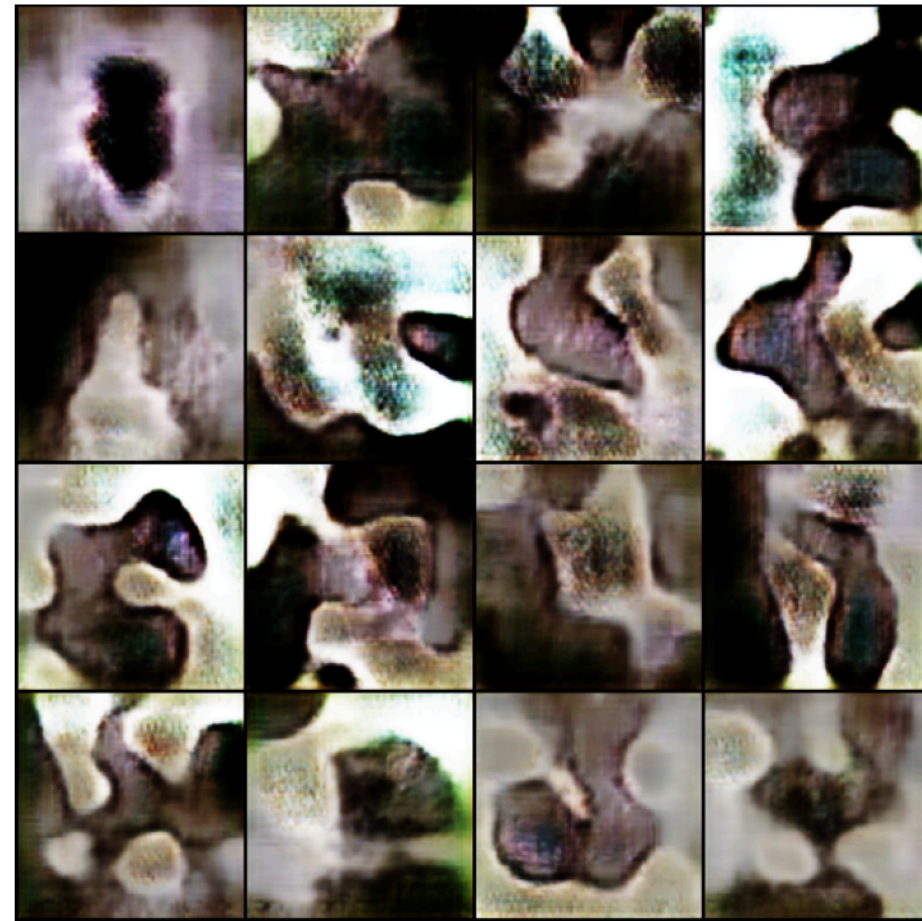
Question: does mixed cats+dogs training improve robustness, or does it dilute cat generation? Same DCGAN architecture.

Extension: same architecture, different training data — 128×128, best of three families

Cats only (DCGAN)
FID=233.0



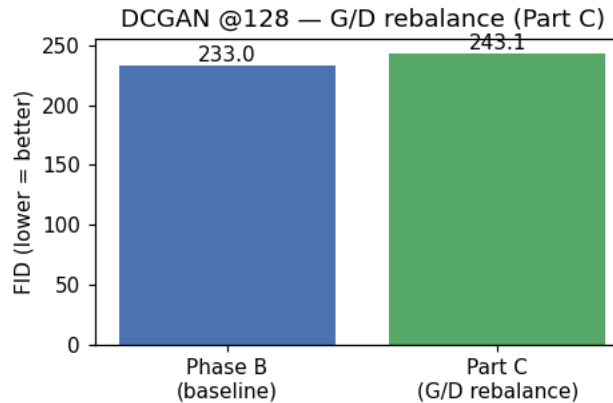
Cats + dogs (DCGAN)
FID=271.0



Part C — G/D rebalance @128

Phase B DCGAN showed a **generator-loss plateau**, suggesting D was too strong. Part C rebalanced G vs D on the same 3000-cat split for 80 epochs.

	Phase B	Part C
<code>lr (G)</code>	1e-4	2e-4
<code>lr_d</code>	2e-4	1e-4
<code>label_smooth</code>	0.1	0.05



FID: 232.964 → 243.147 — rebalance did **not** improve on Phase B

Part C — Baseline vs rebalance (eval samples)

DCGAN eval samples — baseline vs G/D rebalance

Phase B
FID=233.0



Part C
FID=243.1

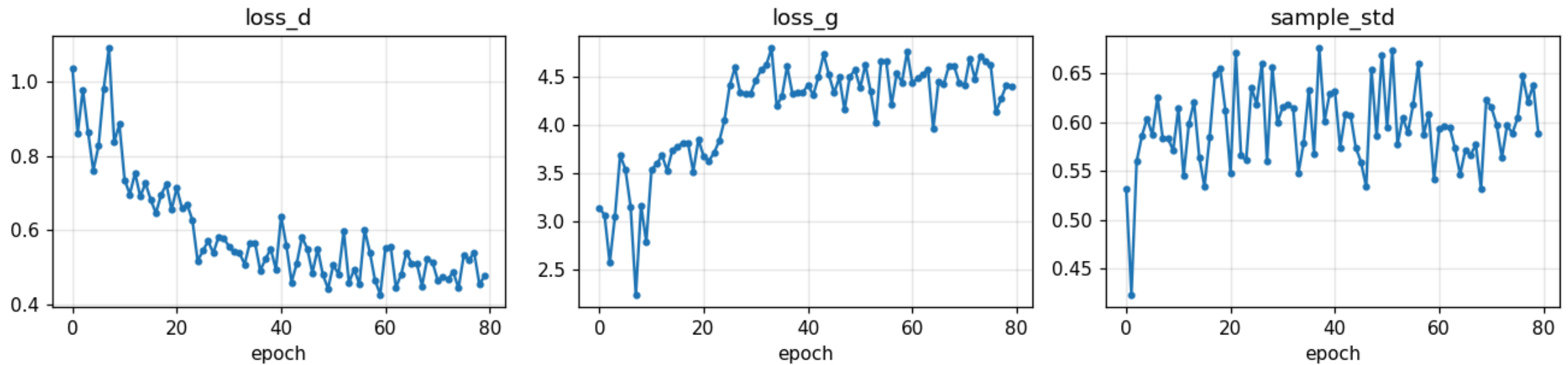


Baseline DCGAN
FID 232.964

Rebalanced DCGAN
FID 243.147

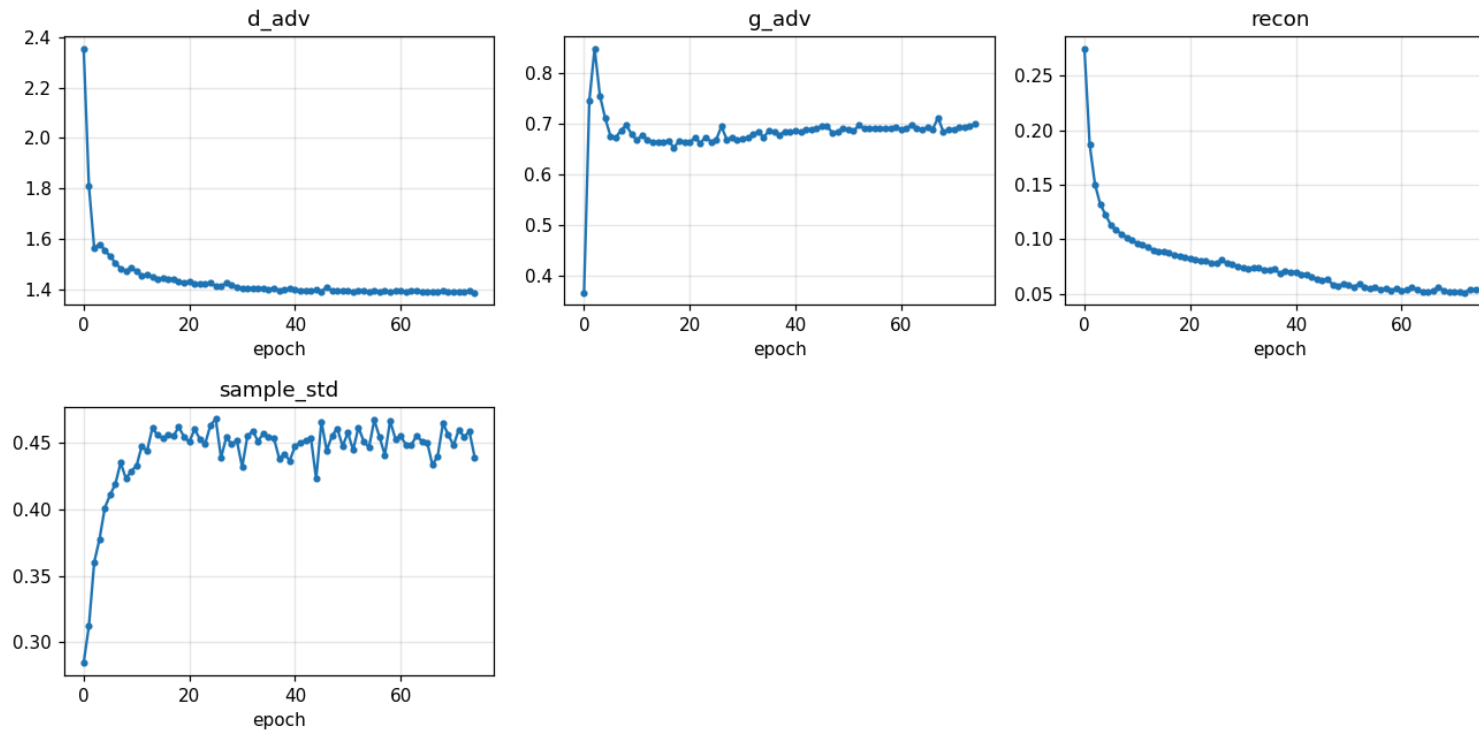
Learning curves — DCGAN

Adversarial losses are diagnostic rather than direct image-quality objectives. The plateau motivated the Part C G/D rebalance.



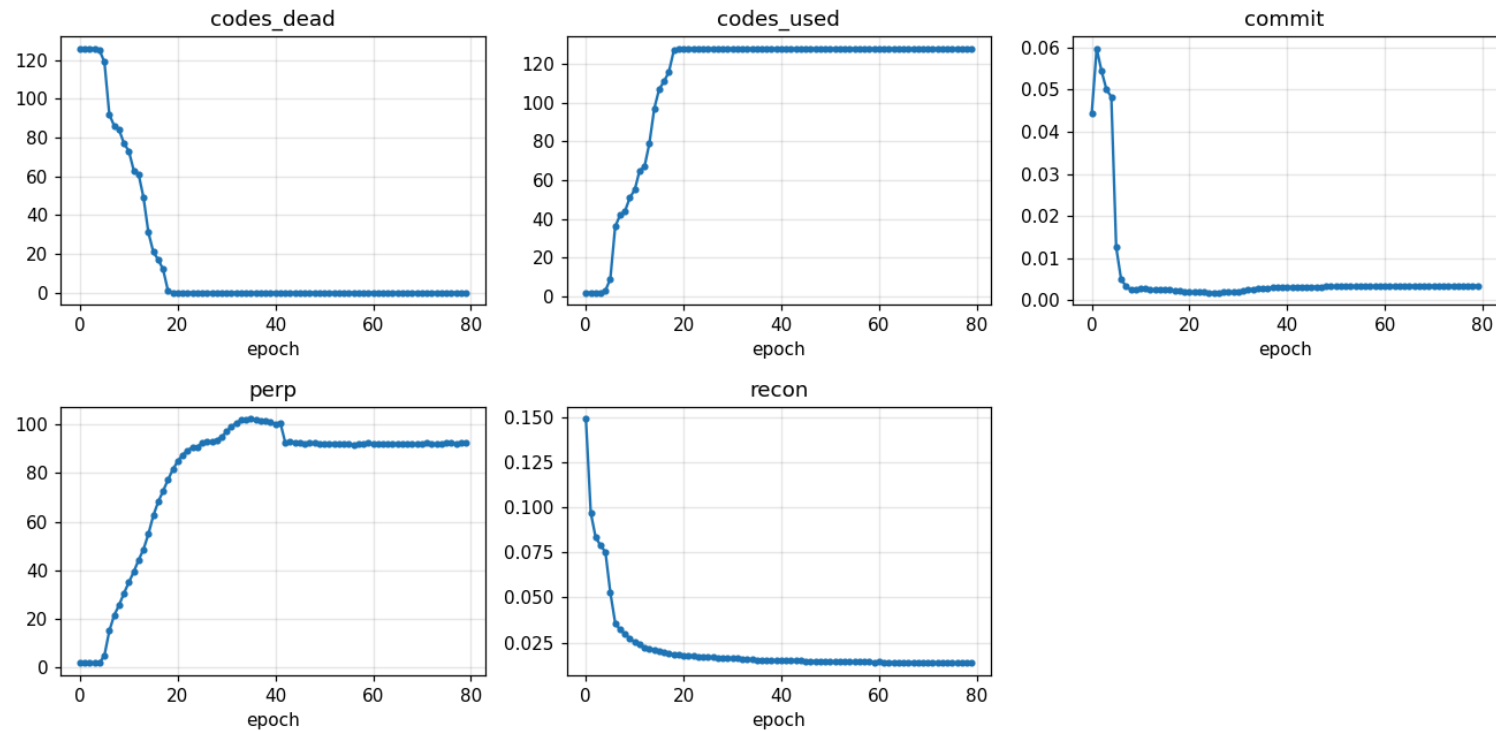
Learning curves — AAE

AAE optimizes reconstruction while an adversarial latent discriminator pushes encoded latents toward the Gaussian prior.



Learning curves — VQ-VAE

VQ-VAE optimizes reconstruction and codebook commitment; low sample diversity suggests empirical code sampling is weak.



Quality diagnostics @128×128

Single-run diagnostics from `eval/quality.json` ; $\mu \pm \sigma$ is over generated samples/features, not repeated seeds.

Model	FID ↓	Sharpness sample $\mu \pm \sigma$	Diversity sample $\mu \pm \sigma$	NN mean
DCGAN	232.964	0.0169±0.0083	18.493±1.736	17.062
AAE	289.050	0.0087±0.0024	16.075±2.154	16.962
VQ-VAE	489.819	0.0178±0.0010	9.590±1.800	17.245

- **Interpretation:** DCGAN has the best FID and highest diversity; VQ-VAE is sharp by this statistic but much less diverse
- Sharpness: Laplacian variance over generated samples
- Diversity: mean pairwise Inception-feature distance
- NN mean: nearest-neighbor distance to training references in Inception space

Conclusions

Phase	Best model	Best cats FID
64×64 refine	DCGAN	287.024
128×128	DCGAN	232.964
128×128 G/D rebalance	DCGAN	243.147

1. **DCGAN** wins across phases (AAE/VQ-VAE higher FID)
2. **64×64 sweep + refine** = fair comparison, **not** good visuals
3. **128×128 + more data** = clear improvement
4. **G/D rebalance** did not beat Phase B — simple LR/smoothing tweak insufficient

Limitations

- Small **custom** nets — not StyleGAN / diffusion
- **FID not comparable** across 64 vs 128 (different resolution & splits)
- **FID-only** — no human study; **VQ-VAE** has no latent interpolation in our pipeline
- No repeated seeded trials, so FID has no confidence interval here
- Cats-vs-dogs extension only for **best** model at 128; Part A used smaller data splits

Thank you

github.com/AdamKaniasty/DeepLearningCatGen